

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A hospital stores the body temperature readings of 12 patients in consecutive memory locations. A patient is considered to have a fever if the temperature value is greater than the fever threshold stored in memory. Develop an 8085 ALP to:
 - Count the number of patients with fever.
 - Find the highest recorded temperature.
 - Store both results in memory.
2. An automated irrigation system stores water levels measured every hour. If the water level falls below the minimum safe level, an alert should be generated. Write an 8085 ALP to:
 - Count the number of low-water-level readings.
 - Determine the lowest water level recorded.
 - Store an alert flag if more than three readings are below the safe level.
3. A library stores the number of books borrowed from ten different sections. The librarian wants to identify the most frequently used section and determine the total number of books borrowed. Write an 8085 ALP to:
 - Calculate the total books borrowed.
 - Find the section with the highest borrowing count.
 - Store both results.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

Load me at

1LXI H, 8050

2MOV B, M

3INX H

4MOV A, M

5MOV D, A

6MVI C, 00

7MVI E, 0CH

8

9LOOP: MOV A, M

10CMP B

11JC SKIP1

12INR C

13

14SKIP1: MOV A, M

15CMP D

16JC SKIP2

17MOV D, A

18

19SKIP2: INX H

20DCR E

21JNZ LOOP

22

23LXI H, 8060

24MOV M, C

25INX H

26MOV M, D

27HLT

DataStackKeypadMemory

Start8050

Address (Hex)	Address	Data
1F72	8050	26
1F73	8051	24
1F74	8052	27
1F75	8053	28
1F76	8054	25
1F77	8055	29
1F78	8056	26
1F79	8057	27
1F7A	8058	24
1F7B	8059	26
1F7C	8060	8
1F7D	8061	29
1F7E	8062	28
1F7F	8063	0

Line No	Assembler Message
0	Program assembled successfully

```

1  ● LXI H,8050H
2  MVI C,0AH
3  MOV A,M
4  MOV E,A
5  MVI B,00H
6
7  LOOP: MOV A,M
8  CPI 30H
9  JNC CHECK
10 INR B
11
12 CHECK: MOV A,M
13 CMP E
14 JNC NEXT
15 MOV E,A
16
17 NEXT: INX H
18 DCR C
19 JNZ LOOP
20
21 MOV A,B
22 STA 8057
23
24 MOV A,E
25 STA 8058
26
27 MOV A,B
28 CPI 04H
29 JC NOALERT
30
31 MVI A,0FFH
32 STA 8059
33 HLT
34
35 NOALERT: MVI A,00H
36 STA 8059
37 HLT

```

Start 8050

Address (Hex)	Address	Data
1F72	8050	28
1F73	8051	35
1F74	8052	40
1F75	8053	18
1F76	8054	25
1F77	8055	15
1F78	8056	35
1F79	8057	20
1F7A	8058	45
1F7B	8059	22
1F7C	8060	0
1F7D	8061	0
1F7E	8062	0
1F7F	8063	0

Line No	Assembler Message
0	Program assembled successfully

Load me at

```
1  LXI H, 8050H
2  MVI C, 0AH
3
4  MOV A,M
5  MOV B,A
6  MOV D,A
7
8  INX H
9  DCR C
10
11 LOOP: MOV A,D
12 ADD M
13 MOV D,A
14
15 MOV A,M
16 CMP B
17 JC NEXT
18
19 MOV B,A
20
21 NEXT: INX H
22 DCR C
23 JNZ LOOP
24
25 MOV A,D
26 STA 8058H
27
28 MOV A,B
29 STA 8059H
30
31 HLT
```

Data

Stack

Keypad

Memory

Start 8050

Address (Hex)	Address	Data
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1F72	8050	5
------	------	---

1F73	8051	8
------	------	---

1F74	8052	4
------	------	---

1F75	8053	9
------	------	---

1F76	8054	7
------	------	---

1F77	8055	3
------	------	---

1F78	8056	6
------	------	---

1F79	8057	2
------	------	---

1F7A	8058	44
------	------	----

1F7B	8059	9
------	------	---

1F7C	8060	0
------	------	---

1F7D	8061	0
------	------	---

1F7E	8062	0
------	------	---

1F7F	8063	0
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Line No	Assembler Message
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0	Program assembled successfully
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